

Technology Skill Development Lab Java

Assignment No.6

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Problem Statement : Write a Java program in which you will declare an abstract class Vehicle inherits this class from two classes car and truck using the method engine in both displays “car has good engine” and “truck has bad engine”. Write a Java program to demonstrate abstraction and inheritance by creating a general vehicle structure and extending it into Car and Truck with different engine behavior. Assign a performance score to each vehicle. Apply a Rule-Based Classification Algorithm to categorize vehicles as High, Medium, or Low Performance. Display the vehicle details along with performance score and category

Program:

```
import java.util.Scanner;

abstract class Vehicle {

    int performanceScore;
    Scanner sc = new Scanner(System.in);

    abstract void engine();

}

class Car extends Vehicle {

    void engine() {

        System.out.print("Enter Car Performance Score: ");
```

```
        performanceScore = sc.nextInt();

        System.out.println("Performance Score: " + performanceScore);
        System.out.println("Car has good engine");

        if (performanceScore >= 80 && performanceScore <= 100) {
            System.out.println("Category: High Performance");
        }
        else if (performanceScore >= 50 && performanceScore < 80) {
            System.out.println("Category: Medium Performance");
        }
        else {
            System.out.println("Category: Low Performance");
        }
    }
}

class Truck extends Vehicle {

    void engine() {

        System.out.print("Enter Truck Performance Score: ");
        performanceScore = sc.nextInt();

        System.out.println("Performance Score: " + performanceScore);
```

```
        System.out.println("Truck has bad engine");

        if (performanceScore >= 80 && performanceScore <= 100) {
            System.out.println("Category: High Performance");
        }
        else if (performanceScore >= 50 && performanceScore < 80) {
            System.out.println("Category: Medium Performance");
        }
        else {
            System.out.println("Category: Low Performance");
        }
    }
}

public class ass6 {

    public static void main(String[] args) {

        Car c1 = new Car();

        System.out.println("----- Car Details -----");
        c1.engine();

        System.out.println();
```

```
        Truck t1 = new Truck();

        System.out.println("----- Truck Details -----");
        t1.engine();
    }
}
```

Output :

```
----- Car Details -----
Enter Car Performance Score: 85
Performance Score: 85
Car has good engine
Category: High Performance

----- Truck Details -----
Enter Truck Performance Score: 77
Performance Score: 77
Truck has bad engine
Category: Medium Performance
PS C:\Users\ADMIN\OneDrive\Desktop\ss\garry\saurabhhhh_TSDL.java>
```

Conclusion: This Java program demonstrates abstraction and inheritance using an abstract Vehicle class and its subclasses Car and Truck. The abstract engine() method is overridden in both classes to accept a performance score from the user and categorize it as High, Medium, or Low Performance. The program also shows how different vehicle types can provide their own implementation of the same abstract method. Overall, the program successfully demonstrates the concepts of abstract classes, method overriding, inheritance, user input, and conditional statements in Java.